UTM UNIVERSITI TEKNOLOGI MALAYSIA
FACULTY OF COMPUTING
UTM Johor Bahru

# PROJECT PART #1

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| Subject | : | Human-Computer Interaction Fundamentals (SCSM2113) |
| Session | : | 2025/2026 Semester 1 |
| Title | : | Project Proposal |
| Duration | : | 1 Week |
| Submission | : | This is a **TEAM ASSIGNMENT** and must be submitted<br>(1) Latest by 14th Nov 2025<br>(2) Post onto the setup platform |

***NOTE TO STUDENTS***

*This semester, your HCI Project 1 will integrate input from an industry partner — DreamEDGE Sdn Bhd, who has proposed a theme on* ***Digital Twin Visualisation Platform for Smart Industry****. While the industry theme focuses on technology and data systems, your HCI task is to explore the human interaction and usability aspects of digital twin systems. You are encouraged to apply HCI design thinking to understand how users interact with such systems, what challenges they face, and how design can make these technologies more intuitive, trustworthy, and effective.*

*This ensures that your project remains grounded in human-centred design, while still contributing meaningfully to the industry theme.*

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**Theme:** Enhancing Human Interaction and Decision-Making in Digital Twin Systems
**Based on Industry Input from:** DreamEDGE – *Digital Twin Visualisation Platform for Smart Industry*

**Context**

DreamEDGE has introduced the concept of Digital Twin systems, i.e. virtual models that mirror real industrial assets to visualise, simulate, and predict their performance. These systems integrate data, sensors, and visualisation technologies to enable real-time monitoring and analytics.

While digital twins are technically advanced, **their success depends heavily on how humans interact with them**. Users such as factory supervisors, engineers, or operators must be able to interpret complex data, trust system feedback, and make timely decisions based on the digital twin interface. This project focuses on the **Human-Computer Interaction (HCI)** aspect, specifically understanding user challenges and designing interactive solutions that enhance the experience, usability, and trustworthiness of digital twin technologies.

**Task Overview**
As an HCI designer, your team will explore the human-centred challenges of digital twin systems and propose design concepts that make them more usable, understandable, and effective.

You are **not required to build a fully functional industrial system**. Still, you should instead **identify real or potential usability and user experience issues** in how digital twin interfaces are designed or used, and propose an interactive solution addressing those issues.

**Possible Directions (Choose one or combine several)**

1. **Usability Challenge:** How can complex sensor or performance data be visualised more intuitively for non-technical users?
2. **Experience Challenge:** How can digital twin dashboards reduce information overload and support user attention, confidence, or focus during monitoring tasks?
3. **Interaction Challenge:** What new interaction methods (e.g., touch, gesture, voice, AR) can make real-time data monitoring and control more effective?
4. **Trust and Transparency:** How might users gain trust and interpret predictive insights or alerts from the system?
5. **Accessibility and Inclusivity:** How can digital twin interfaces be designed for diverse user groups (e.g., operators, trainees, managers)?

**<u>Submission 1: Project Proposal</u>**

Your proposal should include the following sections:

**Problem**

- Identify the usability or user experience problems related to digital twin interfaces or monitoring systems.
- Describe at least **three (3)** problems and classify them under *usability* and *user experience* goals.
- Support your analysis with relevant references or examples (e.g., images of dashboards, existing digital twin systems).
- Ensure your problem reflects real or emerging challenges in industry contexts and relates to at least one UN Sustainable Development Goal (SDG), such as:
    - SDG 9: Industry, Innovation, and Infrastructure
    - SDG 8: Decent Work and Economic Growth
    - SDG 12: Responsible Consumption and Production

**Proposed Solution**

- Propose an interactive system, interface, or product design that addresses the identified usability and UX problems.
- Describe what it is (e.g., smart dashboard, AR-based control panel, adaptive alert system).
- Explain how the proposed design supports better interaction, comprehension, and decision-making.

**Target Users**

- Identify at least **three (3)** user groups relevant to your chosen context (e.g., maintenance engineers, factory supervisors, system trainees).
- Explain how each group would benefit from your proposed system.

**Reference Slides Teaching:**

Topic 3 – Conceptualising Interaction

*Rubric:*

| **Item** | **Minimal (1)** | **Moderate (2)** | **Good (3)** | **Excellent (4)** | **Weight** | **Score** |
|---|---|---|---|---|---|---|
| Problem | Problems identified are not usability or UX problems | Weak explanation of identified usability and UX problems | Clear explanation of identified usability and UX problems | Excellent understanding of the problem space and its UX implications | 2 | 8 |
| Illustration | 1 related image provided | 2 related images provided | 3 related images provided | Clear, referenced illustrations supporting each problem | 1 | 4 |
| Proposed Solution | General idea unrelated to UX | Addresses some UX problems | Addresses most UX problems | Strongly relates and solves all identified UX issues | 2 | 8 |
| Target Users | Users not distinguished | Users are identified but weakly linked to the design | Users distinguished with partial benefit explanation | Clear distinction and strong justification of benefits to each group | 1 | 4 |

*24/24 = 5%*